Test_Case __Wbt_001

| Field | Description |
|---|---|
| Taste Case ID | TC_Wbt_001 |
| Title | Verify successful login authentication |
| Description | Verify the execution of the login function when valid username and password are entered. |
| Preconditions | 1. Flask server is running.<br>2. MongoDB Atlas is connected.<br>3. User account exists in the database.<br>4. Login page is accessible. |
| Test Steps | 1. Open the Login page.<br>2. Enter a valid username.<br>3. Enter a valid password.<br>4. Click the **Login** button.<br>5. Verify the execution of the `login_user()` function |
| Test Data | Username: test_user<br>Password: [REDACTED TEST PASSWORD] |
| Expected Result | The system authenticates the user, generates a JWT token, and redirects the user to the dashboard. |
| Actual Result | User authenticated successfully, JWT token generated, and dashboard displayed. |
| Status | Pass |
| Postconditions | User session is created successfully, JWT token is stored, and the dashboard is accessible. |

Test_Case_002

| Field | Description |
|---|---|
| Taste Case ID | TC_Wbt_002 |
| Description | Test if users can search for an existing product |
| Title | Verify invalid password authentication |
| Preconditions | 1. Flask server is running.<br>2. MongoDB Atlas is connected.<br>3. User account exists in the database.<br>4. Login page is accessible. |
| Test Steps | 1. Open the Login page.<br>2. Enter a valid username.<br>3. Enter an incorrect password.<br>4. Click the **Login** button.<br>5. Verify the execution of the invalid password branch in the `login_user()` function. |
| Test Data | Username: test_user<br>Password: wrong123 |
| Expected Result | The system should reject the login request and display an **"Invalid Credentials"** message without generating a JWT token. |
| Actual Result | Login request rejected, error message displayed, and no JWT token generated. |
| Status | Pass |
| Postconditions | User remains on the login page, no session is created, and access to the dashboard is denied. |

TC_003

| Field | Description |
|---|---|
| Taste Case ID | TC_Wbt_003 |
| Title | Verify login validation for empty credentials |
| Description | Verify that the login function validates empty username and password fields and prevents authentication. |
| Preconditions | 1. Flask server is running.<br>2. MongoDB Atlas is connected.<br>3. Login page is accessible.<br>4. Input validation is enabled. |
| Test Steps | 1. Open the Login page.<br>2. Leave the username field empty.<br>3. Leave the password field empty.<br>4. Click the **Login** button.<br>5. Verify the validation logic in the `login_user()` function. |
| Test Data | Username: Empty<br>Password: Empty |
| Expected Result | The system should display validation messages and should not execute the authentication process. |
| Actual Result | Validation messages displayed successfully and authentication process was not executed. |
| Status | Pass |
| Postconditions | User remains on the login page, no session is created, no JWT token is generated, and the user is prompted to enter valid credentials. |

TC_004

| Field | Description |
|---|---|
| Taste Case ID | TC_Wbt_004 |
| Title | Verify product retrieval from MongoDB database |
| Description | Verify that the `get_products()` function executes all statements and retrieves product records successfully from the MongoDB database. |
| Preconditions | 1. Flask server is running.<br>2. MongoDB Atlas is connected.<br>3. Product records are available in the database.<br>4. User is logged into the application. |
| Test Steps | 1. Login to the application.<br>2. Navigate to the Products page.<br>3. Execute the `get_products()` function.<br>4. Verify that product data is fetched from MongoDB.<br>5. Check whether all products are displayed. |
| Test Data | Products Collection: Cricket Bat, Cricket Ball, Gloves, Helmet |
| Expected Result | All product records should be retrieved successfully from MongoDB and displayed on the Products page. |
| Actual Result | Product records were retrieved successfully and displayed correctly |
| Status | Pass |
| Postconditions | Product list is loaded successfully, and users can browse all available products. |

TC_005

| Field | Description |
|---|---|
| Taste Case ID | TC_Wbt_005 |
| Title | Verify product search functionality |
| Description | Verify that the search_product() function correclty executes the branch for valid product search and returns the matching product details |
| Preconditions | 1. Flask server is running.<br>2. MongoDB Atlas is connected.<br>3. Products are available in the database.<br>4. User is logged into the application.<br>5. Search feature is enabled. |
| Test Steps | 1. Login to the application.<br>2. Navigate to the Products page.<br>3. Enter **"Cricket Bat"** in the search box.<br>4. Click the **Search** button.<br>5. Verify the execution of the `search_product()` function and the search result. |
| Test Data | Search Keyword: Cricket Bat |
| Expected Result | The system should retrieve and display the matching product details from the database. |
| Actual Result | The matching product was retrieved successfully and displayed on the Products page. |
| Status | Pass |
| Postconditions | The searched product is displayed successfully, allowing the user to view its details or add it to the cart. |